Codroid SDK

1.0.0

Doxygen 1.9.1

Chapter 1

Codroid SDK

Codroid SDK C++ TCP movJ/movL/movC IO

1.1

- ** ** Asio nlohmann/json third_party
- ** **
- ** ** SDK
  - ** ** (mm)
  - ** ** (deg) 3
- ** ** Linux (GCC) Windows (Visual Studio / MSVC)

1.2

- include/Codroid/ - SDK
- src/ - SDK
- third_party/ - Asio Standalone, nlohmann/json
- examples/ - IO
- build_linux.sh - Linux
- build_msvc.bat - Windows MSVC

1.3

1.3.1 1. Linux (Ubuntu)

```
chmod +x build_linux.sh
./build_linux.sh
```

- build_linux/libCodroid.so
- build_linux/ move_test

```
cd build_linux
export LD_LIBRARY_PATH=.:$LD_LIBRARY_PATH
./move_test
```

1.3.2 2. Windows (Visual Studio / MSVC)

Windows CMake Visual Studio 2019/2022

```
build_msvc.bat
```

- build_msvc/Debug
- build_msvc/Release

Codroid.dll ( ), Codroid.lib ( )

1.4

SDK PIMPL C++ SDK

1. (Include Directories)

Include Path

- CodroidSDK/include
- CodroidSDK/third_party (asio.hpp)

2. (Preprocessor Definitions)

- ASIO_STANDALONE (Asio Boost)

3.Windows

- Codroid.lib
- /utf-8
- Codroid.dll .exe

1.5

```
{c++}
#include <iostream>
#include <vector>
#include ”Codroid/CodroidControlInterface.h”
int main() {
    Codroid::CodroidControlInterface robot;
    // 1.      (     9001)
    if (robot.connect(”192.168.1.136”, 9001)) {
        std::cout « ”Connected to Codroid Robot!” « std::endl;
        // 2.          ( : mm, deg)
        //
        std::vector<double> currentPos = robot.getCartesianPosition();
        std::cout « ”Current X: ” « currentPos[0] « ” mm” « std::endl;
        // 3.
        //         [j1..j6]
        std::vector<double> home = {0.0, 0.0, 90.0, 0.0, 90.0, 0.0};
        robot.movJ(home, 50.0, 100.0);
        // 4.
        robot.disconnect();
    } else {
        std::cerr « ”Connection failed!” « std::endl;
    }
    return 0;
}
```

1.6

1. 1 / Wait 1
2. SDK coor tool Bug
3. sendCommand std::mutex
4. joint SDK Degrees SDK 1rad 57.295deg

Chapter 2

2.1

:

Chapter 3

3.1

:

Chapter 4

4.1

:

Chapter 5

5.1 Codroid::CodroidControlInterface

Public

- CodroidControlInterface ()
- ~CodroidControlInterface ()
- CodroidControlInterface (const CodroidControlInterface &)=delete
- CodroidControlInterface & operator= (const CodroidControlInterface &)=delete
- bool connect (const std::string &ip, int port=9001)
- void disconnect ()
- Response sendCommand (const std::string &type, const json &data, int id)
- Response runScript (const std::string &script, int id=1)
- Response enterRemoteScriptMode (int id=1)
- Response runProject (const std::string &projectid, int id=1)
  ID
- Response runProjectByIndex (int index, int id=1)
- Response runStep (const std::string &projectid, int id=1)
- Response pauseProject (int id=1)
- Response resumeProject (int id=1)
- Response stopProject (int id=1)
- Response setStartLine (int startline, int id=1)
- Response clearStartLine (int id=1)

- Response getGlobalVars (int id=1)
- Response saveGlobalVars (const std::map< std::string, Variable > &vars, int id=1)
- Response removeGlobalVars (const std::vector< std::string > &varNames, int id=1)
- Response getProjectVar (int id=1)
- Response RS485init (int baudrate, RS485StopBits stopBit=RS485StopBits::One, int dataBit=8, RS485Parity parity=RS485Parity::None, int id=1)

  RS485
- Response RS485flush (int id=1)

  RS485
- Response RS485read (int length, int timeout=5000, int id=1)

  RS485
- Response RS485write (const std::vector< uint8_t > &data, int id=1)

  RS485
- std::vector< double > forwardKinematics (const FKParams &params, int id=1)
- std::vector< double > inverseKinematics (const IKParams &params, int id=1)
- std::vector< double > calculateRelativePose (const RelativePoseParams &params, int id=1)
- Response jog (const JogParams &params, int id=1)
- Response stopJog (int id=1)
- Response jogHeartbeat (int id=1)

  , 500ms
- Response moveTo (const MoveToParams &params, int id=1)
- Response moveToHeartbeat (int id=1)

  moveTo , 500ms
- Response setManualSpeedRate (int speed, int id=1)
- Response setAutoSpeedRate (int speed, int id=1)
- Response move (const std::vector< MoveInstruction > &path, int id=1)
- Response movJ (const MoveInstruction &inst, int id=1)
- Response movJ (const std::vector< double > &jp, double speed, double acc, int id=1)
- Response movL (const MoveInstruction &inst, int id=1)
- Response movL (const std::vector< double > &cp, double speed, double acc, const std::vector< double > &coor={}, const std::vector< double > &tool={}, int id=1)
- Response movC (const MoveInstruction &inst, int id=1)

- Response movCircle (const MoveInstruction &inst, int id=1)
- Response pauseMove (int id=1)
- Response resumeMove (int id=1)
- Response stopMove (int id=1)
- Response switchOn (int id=1)
- Response switchOff (int id=1)
- Response toManual (int id=1)
- Response toAuto (int id=1)
- Response toRemote (int id=1)
- Response toSimulation (int id=1)
- Response toActual (int id=1)
- Response startDrag (int id=1)
- Response stopDrag (int id=1)
- Response clearError (int id=1)
- std::vector< IOInfo > getIOValues (const std::vector< IOInfo > &queryList, int id=1)

  IO
- int getDI (int port, int id=1)
- int getDO (int port, int id=1)
- double getAI (int port, int id=1)
- double getAO (int port, int id=1)
- Response setIOValue (const std::string &type, int port, double value, int id=1)

  IO
- Response setDO (int port, int value, int id=1)
- Response setAO (int port, double value, int id=1)
- std::vector< RegisterInfo > getRegisterValues (const std::vector< int > &addresses, int id=1)
- double getRegisterValue (const int address, int id=1)
- Response setRegisterValue (int address, double value, int id=1)
- Response setExtendArrayType (int index, ExtendArrayType type, int id=1)

- Response removeExtendArray (int index, int id=1)
- Response startDataPush (const std::string &ip, int port, int duration, int id=1)
  IP
- Response stopDataPush (int id=1)
- Response startControl (int duration, int startBuffer=5, int filterTypeint=1, int id=1)
- Response stopControl (int id=1)
- Response setCollisionSensitivity (int level, int id=1)
- Response setPayload (int payloadId, int id=1)

Public

- static void printResponse (const Codroid::Response &resp)

5.1.1

5.1.1.1 calculateRelativePose()

std::vector<double> Codroid::CodroidControlInterface::calculateRelativePose (
const RelativePoseParams & params,
int id = 1)

params	
id	ID

5.1.1.2 clearError()

Response Codroid::CodroidControlInterface::clearError (
int id = 1)

id	ID

5.1.1.3 clearStartLine()

Response Codroid::CodroidControlInterface::clearStartLine (
    int id = 1)

id	ID

5.1.1.4 connect()

bool Codroid::CodroidControlInterface::connect (
    const std::string & ip,
    int port = 9001)

ip	IP
port	TCP (9001)

true

5.1.1.5 enterRemoteScriptMode()

Response Codroid::CodroidControlInterface::enterRemoteScriptMode (
    int id = 1)

id	ID

5.1.1.6 forwardKinematics()

std::vector<double> Codroid::CodroidControlInterface::forwardKinematics (
const FKParams & params,
int id = 1)

params	
id	ID

5.1.1.7 getAI()

double Codroid::CodroidControlInterface::getAI (
int port,
int id = 1)

port	
id	ID

5.1.1.8 getAO()

double Codroid::CodroidControlInterface::getAO (
int port,
int id = 1)

port	
id	ID

5.1.1.9 getDI()

int Codroid::CodroidControlInterface::getDI (
int port,
int id = 1)

port	
id	ID

5.1.1.10 getDO()

int Codroid::CodroidControlInterface::getDO (
int port,
int id = 1)

port	
id	ID

5.1.1.11 getGlobalVars()

Response Codroid::CodroidControlInterface::getGlobalVars (
    int id = 1)

id	ID

5.1.1.12 getIOValues()

std::vector<IOInfo> Codroid::CodroidControlInterface::getIOValues (
    const std::vector< IOInfo > & queryList,
    int id = 1)

IO

queryList	IO
id	ID

IO IOInfo

5.1.1.13 getProjectVar()

Response Codroid::CodroidControlInterface::getProjectVar (
    int id = 1)

id	ID

5.1.1.14 getRegisterValue()

double Codroid::CodroidControlInterface::getRegisterValue (
const int address,
int id = 1)

address	
id	ID

5.1.1.15 getRegisterValues()

std::vector<RegisterInfo> Codroid::CodroidControlInterface::getRegisterValues (
const std::vector< int > & addresses,
int id = 1)

addresses	
id	ID

RegisterInfo

5.1.1.16 inverseKinematics()

std::vector<double> Codroid::CodroidControlInterface::inverseKinematics (
const IKParams & params,
int id = 1)

params	
id	ID

5.1.1.17 jog()

Response Codroid::CodroidControlInterface::jog (
const JogParams & params,
int id = 1)

params	
id	ID

5.1.1.18 jogHeartbeat()

Response Codroid::CodroidControlInterface::jogHeartbeat (
int id = 1)

, 500ms

id	ID

5.1.1.19 movC()

Response Codroid::CodroidControlInterface::movC (
const MoveInstruction & inst,
int id = 1)

inst	
id	ID

5.1.1.20 movCircle()

Response Codroid::CodroidControlInterface::movCircle (
const MoveInstruction & inst,
int id = 1)

inst	
id	ID

5.1.1.21 move()

Response Codroid::CodroidControlInterface::move (
const std::vector< MoveInstruction > & path,
int id = 1)

path	
id	ID

5.1.1.22 moveTo()

Response Codroid::CodroidControlInterface::moveTo (
const MoveToParams & params,
int id = 1)

params	
id	ID

5.1.1.23 moveToHeartbeat()

Response Codroid::CodroidControlInterface::moveToHeartbeat (
int id = 1)

moveTo , 500ms

id	ID

5.1.1.24 movJ() [1/2]

Response Codroid::CodroidControlInterface::movJ (
const MoveInstruction & inst,
int id = 1)

inst	
id	ID

5.1.1.25 movJ() [2/2]

Response Codroid::CodroidControlInterface::movJ (
const std::vector< double > & jp,
double speed,
double acc,
int id = 1)

jp	
speed	
acc	
id	ID

5.1.1.26 movL() [1/2]

Response Codroid::CodroidControlInterface::movL (
const MoveInstruction & inst,
int id = 1)

inst	
id	ID

5.1.1.27 movL() [2/2]

Response Codroid::CodroidControlInterface::movL (
const std::vector< double > & cp,
double speed,
double acc,
const std::vector< double > & coor = {},
const std::vector< double > & tool = {},
int id = 1)

cp	
speed	
acc	
coor	
tool	
id	ID

5.1.1.28 pauseMove()

Response Codroid::CodroidControlInterface::pauseMove (
int id = 1)

id	ID

5.1.1.29 pauseProject()

Response Codroid::CodroidControlInterface::pauseProject (
int id = 1)

id	ID

5.1.1.30 printResponse()

static void Codroid::CodroidControlInterface::printResponse (
const Codroid::Response & resp) [static]

action	
resp	

5.1.1.31 removeExtendArray()

Response Codroid::CodroidControlInterface::removeExtendArray (
int index,
int id = 1)

index	
id	ID

5.1.1.32 removeGlobalVars()

Response Codroid::CodroidControlInterface::removeGlobalVars (
const std::vector< std::string > & varNames,
int id = 1)

varNames	
id	ID

5.1.1.33 resumeMove()

Response Codroid::CodroidControlInterface::resumeMove (
int id = 1)

id	ID

5.1.1.34 resumeProject()

Response Codroid::CodroidControlInterface::resumeProject (
int id = 1)

id	ID

5.1.1.35 RS485flush()

Response Codroid::CodroidControlInterface::RS485flush (
int id = 1)

RS485

id	ID

5.1.1.36 RS485init()

Response Codroid::CodroidControlInterface::RS485init (
int baudrate,
RS485StopBits stopBit = RS485StopBits::One,
int dataBit = 8,
RS485Parity parity = RS485Parity::None,
int id = 1)

RS485

baudrate	RS485
stopBit	RS485
dataBit	RS485
parity	RS485
id	ID

5.1.1.37 RS485read()

Response Codroid::CodroidControlInterface::RS485read (
int length,
int timeout = 5000,
int id = 1)

RS485

length	
timeout	
id	ID

5.1.1.38 RS485write()

Response Codroid::CodroidControlInterface::RS485write (
const std::vector< uint8_t > & data,
int id = 1)

RS485

data	
id	ID

5.1.1.39 runProject()

Response Codroid::CodroidControlInterface::runProject (
const std::string & projectid,
int id = 1)

ID

projectid	ID
id	ID

5.1.1.40 runProjectByIndex()

Response Codroid::CodroidControlInterface::runProjectByIndex (
int index,
int id = 1)

index	
id	ID

5.1.1.41 runScript()

Response Codroid::CodroidControlInterface::runScript (
const std::string & script,
int id = 1)

script	
id	ID

5.1.1.42 runStep()

Response Codroid::CodroidControlInterface::runStep (
const std::string & projectid,
int id = 1)

projectid	ID
id	ID

5.1.1.43 saveGlobalVars()

Response Codroid::CodroidControlInterface::saveGlobalVars (
const std::map< std::string, Variable > & vars,
int id = 1)

vars	
id	ID

5.1.1.44 sendCommand()

Response Codroid::CodroidControlInterface::sendCommand (
const std::string & type,
const json & data,
int id)

type	"runScript" "switchOn"
data	JSON
id	ID 1

5.1.1.45 setAO()

Response Codroid::CodroidControlInterface::setAO (
int port,
double value,
int id = 1)

port	
value	
id	ID

5.1.1.46 setAutoSpeedRate()

Response Codroid::CodroidControlInterface::setAutoSpeedRate (
int speed,
int id = 1)

speed	0-100
id	ID

5.1.1.47 setCollisionSensitivity()

Response Codroid::CodroidControlInterface::setCollisionSensitivity (
int level,
int id = 1)

level	0-5
id	ID

5.1.1.48 setDO()

Response Codroid::CodroidControlInterface::setDO (
int port,
int value,
int id = 1)

port	
value	
id	ID

5.1.1.49 setExtendArrayType()

Response Codroid::CodroidControlInterface::setExtendArrayType (
int index,
ExtendArrayType type,
int id = 1)

index	
type	
id	ID

5.1.1.50 setIOValue()

Response Codroid::CodroidControlInterface::setIOValue (
const std::string & type,
int port,
double value,
int id = 1)

IO

type	IO "DI" "DO" "AI" "↩ AO"
port	
value	
id	ID

5.1.1.51 setManualSpeedRate()

Response Codroid::CodroidControlInterface::setManualSpeedRate (
int speed,
int id = 1)

speed	0-100
id	ID

5.1.1.52 setPayload()

Response Codroid::CodroidControlInterface::setPayload (
int payloadId,
int id = 1)

payloadId	ID
id	ID

5.1.1.53 setRegisterValue()

Response Codroid::CodroidControlInterface::setRegisterValue (
int address,
double value,
int id = 1)

address	
value	
id	ID

5.1.1.54 setStartLine()

Response Codroid::CodroidControlInterface::setStartLine (
int startline,
int id = 1)

startline	
id	ID

5.1.1.55 startControl()

Response Codroid::CodroidControlInterface::startControl (
int duration,
int startBuffer = 5,
int filterTypeint = 1,
int id = 1)

duration	1000/
startBuffer	
filterTypeint	0- 1-
id	ID

5.1.1.56 startDataPush()

Response Codroid::CodroidControlInterface::startDataPush (
const std::string & ip,
int port,
int duration,
int id = 1)

IP

ip	IP
port	
duration	(1000/)
id	ID

5.1.1.57 startDrag()

Response Codroid::CodroidControlInterface::startDrag (
int id = 1)

id	ID

5.1.1.58 stopControl()

Response Codroid::CodroidControlInterface::stopControl (
int id = 1)

id	ID

5.1.1.59 stopDataPush()

Response Codroid::CodroidControlInterface::stopDataPush (
int id = 1)

id	ID

5.1.1.60 stopDrag()

Response Codroid::CodroidControlInterface::stopDrag (
int id = 1)

id	ID

5.1.1.61 stopJog()

Response Codroid::CodroidControlInterface::stopJog (
int id = 1)

id	ID

5.1.1.62 stopMove()

Response Codroid::CodroidControlInterface::stopMove (
int id = 1)

id	ID

5.1.1.63 stopProject()

Response Codroid::CodroidControlInterface::stopProject (
int id = 1)

id	ID

5.1.1.64 switchOff()

Response Codroid::CodroidControlInterface::switchOff (
int id = 1)

id	ID

5.1.1.65 switchOn()

Response Codroid::CodroidControlInterface::switchOn (
int id = 1)

id	ID

5.1.1.66 toActual()

Response Codroid::CodroidControlInterface::toActual (
int id = 1)

id	ID

5.1.1.67 toAuto()

Response Codroid::CodroidControlInterface::toAuto (
int id = 1)

id	ID

5.1.1.68 toManual()

Response Codroid::CodroidControlInterface::toManual (
int id = 1)

id	ID

5.1.1.69 toRemote()

Response Codroid::CodroidControlInterface::toRemote (
int id = 1)

id	ID

5.1.1.70 toSimulation()

Response Codroid::CodroidControlInterface::toSimulation (
int id = 1)

id	ID

:

- include/Codroid/CodroidControlInterface.h

5.2 Codroid::CodroidException

Codroid SDK

#include <CodroidDefine.h>

Public

- CodroidException (const std::string &message)

5.2.1

Codroid SDK

:

- include/Codroid/CodroidDefine.h

5.3 Codroid::CodroidSubscriber

TCP

#include <CodroidSubscriber.h>

Public

- CodroidSubscriber ()
- ~CodroidSubscriber ()
- bool connect (const std::string &ip, int port=9001)
- void disconnect ()
- bool subscribe (const std::string &topic, int cycle=0)
- void setStatusCallback (StatusCallback cb)
- void setPostureCallback (Codroid::PostureCallback cb)
- void setVarCallback (VarCallback cb)
- void setLogCallback (LogCallback cb)

5.3.1

TCP

5.3.2

5.3.2.1 connect()

```
bool Codroid::CodroidSubscriber::connect (
        const std::string & ip,
        int port = 9001 )
```

ip	IP
port	TCP (9001)

true false

5.3.2.2 setLogCallback()

void Codroid::CodroidSubscriber::setLogCallback (
LogCallback cb) [inline]

cb	

5.3.2.3 setPostureCallback()

void Codroid::CodroidSubscriber::setPostureCallback (
Codroid::PostureCallback cb) [inline]

cb	

5.3.2.4 setStatusCallback()

void Codroid::CodroidSubscriber::setStatusCallback (
StatusCallback cb) [inline]

cb	

5.3.2.5 setVarCallback()

void Codroid::CodroidSubscriber::setVarCallback (
VarCallback cb) [inline]

cb	

5.3.2.6 subscribe()

bool Codroid::CodroidSubscriber::subscribe (
const std::string & topic,
int cycle = 0)

15.1

topic	("RobotStatus", "RobotPosture", "Log")
cycle	(0)

true

:

- include/Codroid/CodroidSubscriber.h

5.4 Codroid::FKParams

#include <CodroidDefine.h>

Public

- FKParams (const std::vector< double > &jointPos)

Public

- std::vector< double > jp
  [ ] [j1...j6], : deg
- std::vector< double > coor
  [ ] ,
- std::vector< double > tool
  [ ] ,
- std::vector< double > ep
  [ ]

5.4.1

:

- include/Codroid/CodroidDefine.h

5.5 Codroid::IKParams

#include <CodroidDefine.h>

Public

- IKParams (const std::vector< double > &cartesianPos)

Public

- std::vector< double > cp
  [ ] , : mm, deg
- std::vector< double > rj
  [ ] , [20,20,20,20,20,20]
- std::vector< double > ep
  [ ]

5.5.1

:

- include/Codroid/CodroidDefine.h

5.6 Codroid::IOInfo

IO

#include <CodroidDefine.h>

Public

- IOInfo (const std::string &t, int p)

Public

- std::string type
  IO
- int port

- double value
  IO

5.6.1

IO

:

- include/Codroid/CodroidDefine.h

5.7 Codroid::JogParams

#include <CodroidDefine.h>

Public

- JogParams (JogMode m, double s, int i, CoorType ct=CoorType::User, int cid=1)

Public

- JogMode mode = JogMode::Line
  1: 2:
- double speed = 0.0
  -1∼1
- int index = 1
  /
- CoorType coorType = CoorType::User

- int coorId = 1
  ID

5.7.1

:

- include/Codroid/CodroidDefine.h

5.8 Codroid::MoveInstruction

#include <CodroidDefine.h>

Public

- MoveType type = MoveType::movJ
- double speed = 60.0
- double acc = 150.0
- double blend = -1.0
- double relativeBlend = -1.0
- int circleNum = 1
- MovePoint targetPoint
- MovePoint middlePoint
- std::vector< double > coor
- std::vector< double > tool

5.8.1

:

- include/Codroid/CodroidDefine.h

5.9 Codroid::MoveToParams

#include <CodroidDefine.h>

Public

- MoveToParams (MoveToType t)
- MoveToParams (MoveToType t, const MoveToTarget &tgt)

Public

- MoveToType type = MoveToType::Home
- MoveToTarget target

  [ ]

5.9.1

:

- include/Codroid/CodroidDefine.h

5.10 Codroid::MoveToTarget

#include <CodroidDefine.h>

Public

- MoveToTarget ()=default

  [ ] [j1..j6]

Public

- static MoveToTarget Joint (const std::vector< double > &j)
- static MoveToTarget Cartesian (const std::vector< double > &c)

Public

- std::vector< double > cp

  [ ] [x,y,z,a,b,c]
- std::vector< double > jp

5.10.1

:

- include/Codroid/CodroidDefine.h

5.11 Codroid::ProjectState

#include <CodroidDefine.h>

Public

- std::string id
  ID
- int state
- bool isStep
- std::map< std::string, int > scriptLines
  ID

5.11.1

:

- include/Codroid/CodroidDefine.h

5.12 Codroid::RegisterInfo

#include <CodroidDefine.h>

Public

- RegisterInfo (int addr, double val)

Public

- int address
- double value

5.12.1

:

- include/Codroid/CodroidDefine.h

5.13 Codroid::RelativePoseParams

#include <CodroidDefine.h>

Public

- RelativePoseParams (const std::vector< double > &p, const std::vector< double > &o, CoorType type)

Public

- std::vector< double > pos
  TCP [x,y,z,a,b,c]
- std::vector< double > offset
  [x,y,z,a,b,c]
- CoorType coorType = CoorType::Tool
  [ ]
- std::vector< double > posCoor
  [ ] TCP
- std::vector< double > coor
  [ ] coorType user

5.13.1

:

- include/Codroid/CodroidDefine.h

5.14 Codroid::Response

SDK

#include <CodroidDefine.h>

Public

- int id
  ID
- std::string ty
- json db
- std::string error_msg

5.14.1

SDK

:

- include/Codroid/CodroidDefine.h

5.15 Codroid::RobotPosture

#include <CodroidDefine.h>

Public

- std::vector< double > joint
  ( )
- std::vector< double > cart
  [x,y,z,a,b,c]

5.15.1

:

- include/Codroid/CodroidDefine.h

5.16 Codroid::RobotStatus

#include <CodroidDefine.h>

Public

- int mode
  0: , 1: , 2:
- int state
  0: , 1: , 2: , 3: , 4:RunTo, 5:
- int isMoving
  0: , 1:
- double moveRate
- double manualMoveRate
- int recoveryState
- bool isSimulation
- int teachingPendant
- int rescueFlag
- int modeSwitch
- int ToolId
- int PayloadId
- int CoordinateId
- int defaultToolId
- int defaultPayloadId
- int defaultUserCoorId
- std::string type
- std::string stateName
- long long timestamp

5.16.1

:

- include/Codroid/CodroidDefine.h

5.17 Codroid::Variable

#include <CodroidDefine.h>

Public

- template<typename T >
  Variable (const T &value, const std::string &note="")

Public

- std::string val
  (JSON)
- std::string nm

5.17.1

:

- include/Codroid/CodroidDefine.h

Chapter 6

6.1 include/Codroid/CodroidControlInterface.h

#include "CodroidDefine.h"
#include <unordered_set>
#include <asio.hpp>
#include <memory>
#include <string>

- class Codroid::CodroidControlInterface

6.1.1

6.2 include/Codroid/CodroidDefine.h

#include <string>
#include <vector>
#include <map>
#include <nlohmann/json.hpp>
#include "CodroidExport.h"
#include <stdexcept>
#include <functional>

- struct Codroid::Response
  SDK
- class Codroid::CodroidException
  Codroid SDK
- struct Codroid::MoveInstruction
- struct Codroid::JogParams
- struct Codroid::RelativePoseParams
- struct Codroid::Variable
- struct Codroid::FKParams
- struct Codroid::IKParams
- struct Codroid::MoveToTarget
- struct Codroid::MoveToParams
- struct Codroid::RobotStatus
- struct Codroid::RobotPosture
- struct Codroid::ProjectState
- struct Codroid::IOInfo
  IO
- struct Codroid::RegisterInfo

- using Codroid::json = nlohmann::json
- using Codroid::TopicCallback = std::function< void(const std::string &, const nlohmann::json &)>

- enum class Codroid::RS485Parity : int { None = 0 , Odd = 1 , Even = 2 }
  RS485
- enum class Codroid::RS485StopBits : int { One = 1 , Two = 2 }
  RS485
- enum class Codroid::CoorType { Tool , User }

- Codroid::NLOHMANN_JSON_SERIALIZE_ENUM (CoorType, { {CoorType::Tool, "tool"}, {CoorType::User, "user"} }) enum class MoveType

- NLOHMANN_JSON_SERIALIZE_ENUM(MoveType, { {MoveType::movJ, "movJ"}, {Move↩Type::movL, "movL"}, {MoveType::movC, "movC"}, {MoveType::movCircle, "movCircle"} }) enum class MoveToType Codroid::NLOHMANN_JSON_SERIALIZE_ENUM (MoveToType, { {MoveToType::Home, 0}, {MoveToType::Safe, 1}, {MoveToType::Candle, 2}, {MoveToType::↩Packing, 3}, {MoveToType::Joint, 4}, {MoveToType::Line, 5}, {MoveToType::ResumePoint, 6} }) enum class ExtendArrayType

- Codroid::NLOHMANN_JSON_SERIALIZE_ENUM (ExtendArrayType, { {ExtendArray↩Type::Bool, "Bool"}, {ExtendArrayType::UInt8, "UInt8"}, {ExtendArrayType::Int8, "Int8"}, {ExtendArrayType::UInt16, "UInt16"}, {ExtendArrayType::Int16, "Int16"}, {ExtendArrayType↩::UInt32, "UInt32"}, {ExtendArrayType::Int32, "Int32"}, {ExtendArrayType::Float32, "Float32"} }) enum class JogMode
- Codroid::NLOHMANN_JSON_SERIALIZE_ENUM (JogMode, { {JogMode::Joint, 1}, {Jog↩Mode::Line, 2} }) struct MovePoint

6.2.1

6.2.2

6.2.2.1 NLOHMANN_JSON_SERIALIZE_ENUM() [1/2]

```
Codroid::NLOHMANN_JSON_SERIALIZE_ENUM (
        JogMode ,
        { {JogMode::Joint, 1}, {JogMode::Line, 2} } )
```

< (:)

< (:mm,)

<

<

6.2.2.2 NLOHMANN_JSON_SERIALIZE_ENUM() [2/2]

```
NLOHMANN_JSON_SERIALIZE_ENUM (MoveType, { {MoveType::movJ, "movJ"}, {MoveType::movL, "movL"},
{MoveType::movC, "movC"}, {MoveType::movCircle, "movCircle"} }) enum class MoveToType Codroid::NLOHMANN↩
_JSON_SERIALIZE_ENUM (
        MoveToType ,
        { {MoveToType::Home, 0}, {MoveToType::Safe, 1}, {MoveToType::Candle, 2}, {MoveToType::Packing,
3}, {MoveToType::Joint, 4}, {MoveToType::Line, 5}, {MoveToType::ResumePoint, 6} } )
```

6.3 include/Codroid/CodroidSubscriber.h

CodroidSubscriber

#include <asio.hpp>
#include <nlohmann/json.hpp>
#include <string>
#include <thread>
#include <atomic>
#include <functional>
#include <vector>
#include <map>
#include <mutex>
#include "CodroidDefine.h"

- class Codroid::CodroidSubscriber

  TCP

- using Codroid::StatusCallback = std::function< void(const RobotStatus &)>
- using Codroid::PostureCallback = std::function< void(const RobotPosture &)>
- using Codroid::VarCallback = std::function< void(const std::map< std::string, std::string > &)>
- using Codroid::LogCallback = std::function< void(const std::string &msg, int level)>

6.3.1

CodroidSubscriber